

Generation Interconnection System Impact Study Report

For Queue Project Q22-2243

Ebensburg Power 138 kV

March 23, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2243. The Interconnection Customer proposes a 191.4 MW battery energy storage facility interconnecting at the Ebensburg Power 138 kV substation in the Pennsylvania area. The request was received on November 30, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 643 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$6,119,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Ebensburg Power / Ft Gibson 138kV L
Contingency: Merret / Seaside 138kV
Loading: 138.9% (Rating: 151 MVA)
DFAX: 100.0%
Pre-project loading: 12.3%
2. Facility: Peabody / Ebensburg Power 138kV
Contingency: Ebensburg Power / Ft Gibson 138kV L
Loading: 101.9% (Rating: 198 MVA)
DFAX: 100.0%
Pre-project loading: 5.1%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Fort Gibson / Ebensburg Power 138kV L	Rebuild line with higher-capacity conductor	\$3,103,000	100% (\$3,103,000)
Peabody / Ebensburg Power 138kV	Reconductor line	\$3,016,000	100% (\$3,016,000)
	Total allocated to project		\$6,119,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.